Bayside Property Analysis - Assignment 2

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AD 571

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Visual 1 - Data Card

The card visual displays the average sale value of Bayside properties and total sum of all properties sale price in Bayside. These were made using the AVERAGE and SUM functions on the sale price field. These values provide a quick reference of the total value of property sale price ($10.6b) and the average sale price ($444.15k). They offer a snapshot of market levels and will adjust in response to filters for property types and time. It is valuable as it gives stakeholders a gauge on the market and benchmark to compare individual sales and market segments.

Visual 2 - Line Chart

The line chart shows the yearly sales count trends, with year on the x-axis and the count of sales on the y-axis. The chart shows a peak in 2004, followed by a sharp decline from that point through to 2010, this is representative of the macroeconomic climate during that time. Sales then showed a modest increase until the COVID-19 decline, followed by a sharp recovery in 2021. The visualization highlights market cycles and historical trends, which is useful for understanding past disruptions, anticipating future trends.

Visual 3 - Pie Chart

The pie chart shows the distribution of property types (commercial, residential, other and mixed) by percentage. The chart shows that commercial sales make up 42.68% and residential sales 31.8%, meaning the majority of properties fall within these two core categories. This visualization shows the segmentation of the market and allows focus on the dominant property types. It can also act as a slicer for some of the other charts, allowing for filtered visuals by property type.

Visual 4 - Combined Line and Bar chart

The chart shows the monthly distribution of the residential property sales count in a bar chart, alongside the average monthly sale price in a line chart. The x-axis represents the months, the left y-axis shows the count of sales, and the right y-axis shows the average sale price. The chart shows that the sales volume stays fairly stable across most months, typically between 1,800-2,100 sales. However, the average sale price fluctuates more, dipping early in the year (February-March), before peaking in May. The combination of the sales volume and average sale price gives the chart a richer understanding of the market behavior throughout the year. It helps identify seasonal patterns, when both high sale volume and high average sale price are occurring, or when average sale prices may be higher or lower regardless of volume. It offers insight into how increased demand may impact average sale prices.

Visual 5 - Scatter Plot

The Scatter plot shows the relationship between property size (sqft) and sale price ($) for residential properties. The scatter plot filters for property sales greater than $0 and less than $2,000,000. It also filters for gross square feet greater than 0 sqft and less than 5,000 sqft. The scatter plot has a randomization function built into it, where it randomly generates a sample of 5% of the data to get a snapshot understanding of the residential population. We can see that although there is a weak positive trend between the variables, there is also a significant spread, showing that other factors - location, type, condition of the property - may also affect pricing beyond size.The visualization shows the market variability, and acts as a simplified tool to understand the expectations of prices for properties of different sizes.

Visual 6 - Density Plot (Made by Area Chart)

The density plot displays the count of residential properties and their sale price distribution. I used the sales price along the x-axis and the count of the sales on the y-axis. The density plot was designed with a few parameters to ensure that it produced the best visualization. The data had to be greater than $0, but less than $2,000,000, and the bin sizes were set to $50,000. The chart shows a cluster of property sales between $80,000 and $1,000,000, peaking around $200,000, before significantly skewing to the right-hand side. The chart helps quantify the composition of the dataset and the residential market, guiding sellers and buyers on their expected sale and purchase price.

Average value of Bayside Property Sales ($)

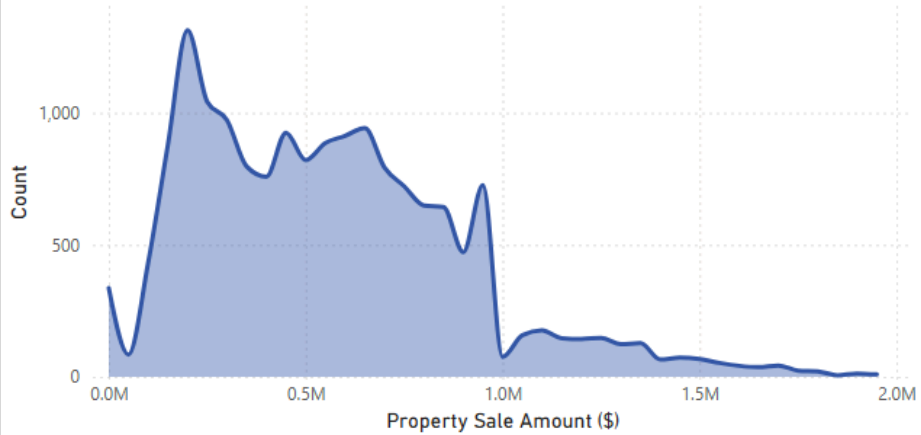
$444.15K

Sum of Bayside Property Sales

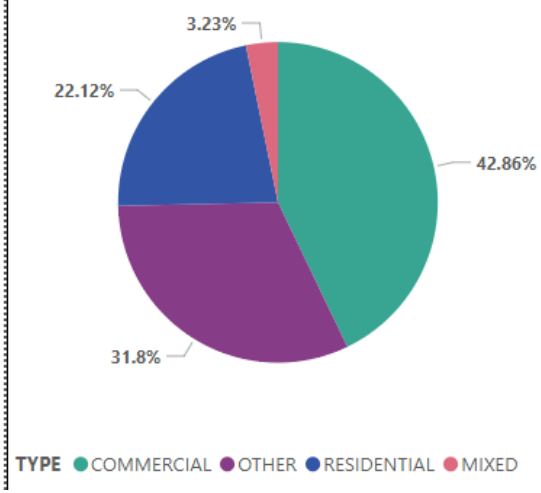
$10.86bn

Bayside Property Sales by Year

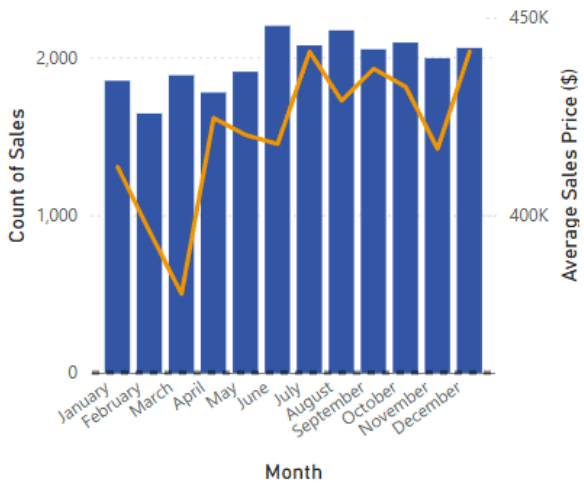
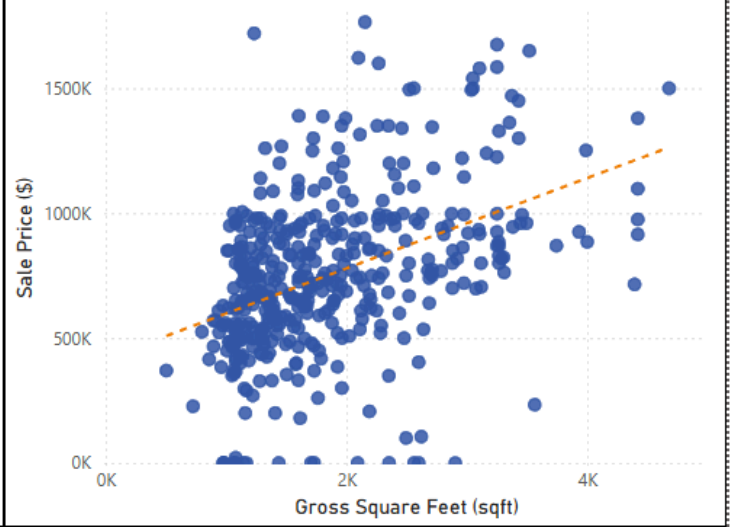
Distribution of Residential Bayside Property Sales



Percentage of Property Types in Bayside



Residential Sales per Month & Avg Sale Price



Distribution of Gross Square Ft & Price - Residential Properties